In March 2014, while I was writing this book, the CMB imprint was discovered by a team assembled by Jamie Bock (Figure 16.10),[30] a cosmologist down the hall from me at Caltech.

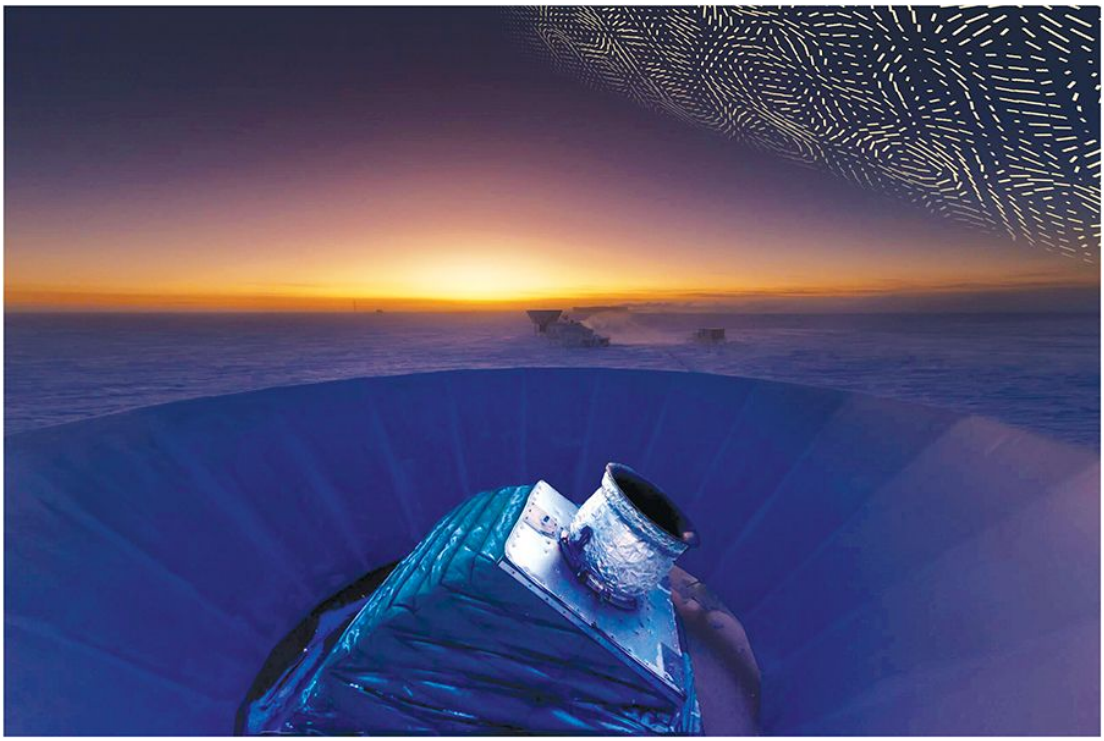


Fig. 16.10. The Bicep2 instrument, built by Jamie Bock's team, that discovered the imprint of primordial gravitational waves. Bicep2, at the South Pole, is here shown at twilight, which occurs only twice a year at the South Pole. It is surrounded by a shield to protect it from radiation from the surrounding ice sheet. The upper right inset shows the measured imprint on the CMB: a polarization pattern. The CMB's electric field points along the short dashed directions.

It was a fantastic discovery, but with a cautionary note: the imprint that Jamie and his team found might possibly be due to something else and not gravitational waves. As this book goes to press, intense efforts are underway to find out for sure.

If the imprint is really due to gravitational waves from the big bang, then this is the type of cosmological discovery that comes along perhaps once every fifty years. It brings us a glimpse of the universe a trillionth of a trillionth of a trillionth of a second after the universe's birth. It